DESCRIPTION OF COURSEWORK

Course Code	EEE316
Course Name	Engineering Management
Lecturer	Ts. Dr. Kang Chia Chao
Academic Session	2023/09
Assessment Title	Project Assignment (Group Work)

A. Introduction/ Situation/ Background Information

Engineering management gives exposure to student in terms of the successful development of the project's procedures of initiation, planning, execution, regulation and closure as well as the guidance of the project team's operations towards achieving all the agreed upon goals within the set scope, time, quality and budget standards. This course seeks to graduate engineers ready to enter professional practice or pursue graduate level studies. This assignment is an opportunity for the student to use engineering management knowledge as a base for solving problems requiring business and analytical skills such as planning, organizing, leading, controlling and cost accounting. Student will be able to work in a wide array of different obstacles and projects.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO1 Demonstrate effective communication on engineering activities in both written and oral forms

CLO3 Engage in retrieval of critical information to propose innovative and creative ideas

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

Project Report

1. You are required to answer all the assignment questions. This is a group assignment. It is important to practice the application of the theoretical information contained in the book or article. You should always remember that in order to obtain maximum marks, it is important that you substantiate the theories with practical application at the same put attention into work originality.
2. These particular aspects will be considered during the assignment marking. By now, you have mastered the course work - this is very important in order to build your confidence as do the assignments.
3. You are required to prepare one softcopy of a typewritten report detailing the answers for the assignment questions. Your report should not more than 25 pages according to report format in section F. Assignment Report Format.

Presentation

1. Presentation must be **not more** than **10-minutes presentation** and **5-minutes Q&A**.
2. Presentation will conduct in physical class.
3. Formal live presentation to the project content clearly as communicated in the presentation rubrics

E. Evaluation Breakdown

No.	Component Title (Report)	Percentage (%)
1.	Report Organisation, Language, Format, References and Citation	10
2.	Product Development, Planning, Creativity, Innovation, Project Risk Management, Test and Refinement, Reflection, Benefit, Recommendation	15
	TOTAL	**25**
No.	**Component Title (Presentation)**	**Percentage (%)**
1.	Content and presentation	4
2.	Delivery of presentation	8
3.	Team work, time management and prototype	3
	TOTAL	**15**

Task(s)

Product Innovation and Development

Teaching aids are tools which facilitate the process of teaching and learning. There are various types of teaching aids, including: traditional teaching aids (e.g. books and blackboard); visual teaching aids (e.g. posters, diagrams, maps); mechanical teaching aids (e.g. audio teaching machines, video-projectors). Modern teaching aids range includes robotics, digital technologies, art and craft, mathematics, literacy resources, developmental products, science equipment, STEM, sporting gear, school furniture, books, play and educational toys.

You are required to come out with a solution and develop a prototype or tool that can be used as teaching aids used to assist lecturers and teachers to more productively manage their learning environments and activities or demonstrations. The scope of your product should limit to the science, technology, engineering and mathematics (STEM) aspects only. The aim is to produce an effective teaching aids used in high school or undergraduate students' learning such as the effect of renewable energy on students' knowledge, attitude toward and behavior of energy saving and carbon reduction or others. You and your team will identify the customer need for the new product, develop a new idea, or improve the existing product performance.

Choose your idea carefully as it must be a manageable product that requires no technological breakthrough (in other words, low cost). Your proposed tool or product should not cost more than RM500.00, the innovative product somewhat novel and provide high impact, and not duplicate any existing products (search the web for possible competitors!). Do some thorough research to identify

competitive products and the solution. You should explain why the existing products do not do the job and convince the examiner that nothing exists that will fill the need you have identified.

This project should be a real problem that you can create a tangible solution to solve. In addition, it should be realistic since your performance in this course will mainly be determined by your ability to provide a real solution to the observed or identified problem. A product prototype is required in the project presentation.

The focus is on product development which involves structured methods, well planned, and the result of the end product is predictable. Using your selected product development process and project management frameworks, write a proposal for the new or innovative product and elevate it to a new level of improvement and transformation.

Requirements (but not limited to):

i. Planning: Brainstorming with team members, identify and delegate tasks, produce Gantt Chart.
ii. Introduction: Establish/find out what the problem is.
iii. System Level Design: Explore possible solutions to overcome the problem
iv. Concept Development: Creative, innovative and/or inventive ideas. Identify your initial material/object.
v. Detail Design: Decide the specific and realistic solution which you will write a proposal about for this project report. Demonstrate the market for the product by identifying the existing products that attempt to meet the need of the community.
vi. Testing and Refinement: Critical design review and production approval.
vii. Design Reflections and Recommendation: Present product design reflection, benefit, advantages and its limitation.
viii. Build Product Prototype: You may use recycle materials or others.
ix. Write a full Project Report: Use the following project template.

F. Assignment Report Format

Assignment Cover Page and Work Declaration



XIAMEN UNIVERSITY MALAYSIA

Course Code :
Course Name :
Lecturer :
Academic Session :
Assessment Title :
Submission Due Date :

Prepared by :

Student ID	Student Name

Date Received :

Feedback from Lecturer:

Mark:



XIAMEN UNIVERSITY MALAYSIA

Own Work Declaration

I/We hereby understand my/our work would be checked for misconduct, and the softcopy would be saved for future comp

I/We hereby confirm that all the references or sources I listed or presented and I/we clearly understand the in intentional or unintentional misconduct.

This work is not made on any work of oth been submitted to any other courses or in

Signature:

Date:

Assignment Report

<Assignment Title>

<Authors>

1. Introduction

<Brief introduction of your work>

1.1. Background study

<Present the background study of your work that includes a review of the area being researched, current information surrounding the issue, previous studies, and relevant history>

1.2. Problem statement

Here you identify the problem of your selected item/material/service and the challenge it faces.

1.3. Objective

<Research objectives describe concisely what the research is trying to achieve. They summarize the accomplishments a researcher wishes to achieve through the project and provide direction to the study>
<Can be in bullet forms or itemized. The keywords such as:

- *i. To develop*
- *ii. To investigate*
- *iii. To improve*
- *iv. To design …*

1.4. Scope of work

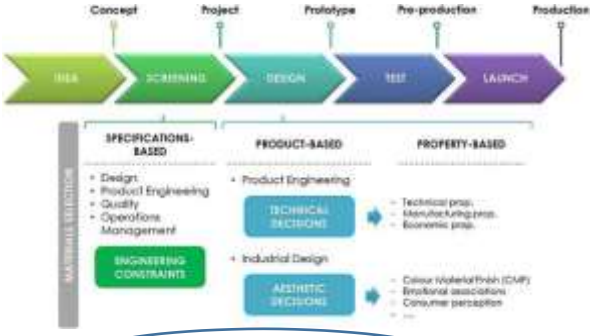
<Which area will be explored in the work and specifies the parameters within the study. You will have to define what the study is going to cover and what it is focusing on>

2. Development of …*<new or enhanced product/service. May refer to your product name>*

<Brief introduction the development of your work>

2.1. Exploration of *<Your product>*

<Can present the concept work and the framework>



Caption Figure 1. Your figure's caption

Table 1. Insert your table title Table Title

2.2. Requirement and Planning

<Brief of project strategy, planning and product requirement>

2.2.1. Project Plan

<You can provide general process flow, detail process flow methodology and its explanation>

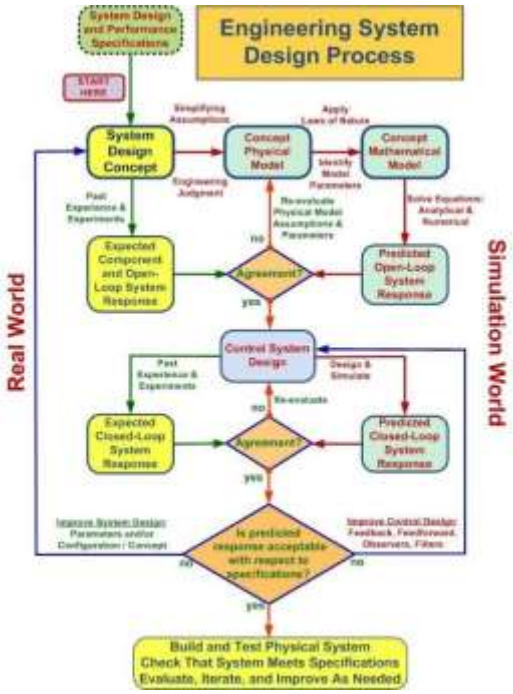


Figure 2. Sample of project process flowchart or can present in any necessary form

2.2.2. Important Milestones

<Present your Project Management using Gantt Chart. Show from the project start in Week 7 to its completion in Week 14 using Excel or free software>

< Show tasks schedule and manpower planning clearly>

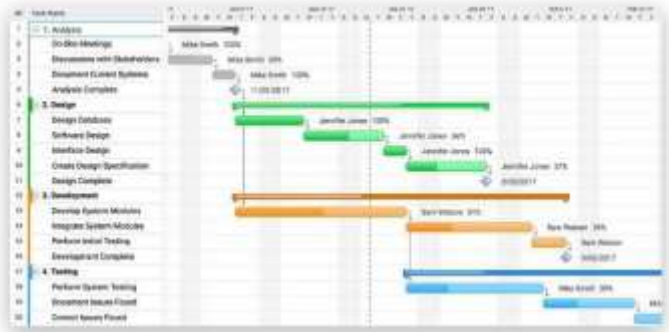


Figure 3. Sample project schedule and manpower planning/organising

2.2.3.Product Requirement

<Present any hardware, software, expert, needed to develop your product>

2.3. Project Risk Management

<Here put a short description of the risk your team and the developed product may have. Provide suggestion or workaround plan>

3. Product Detailed Design

<Brief of design methodology>

3.1. Product Architecture

<Here specify your design architecture and detail materials you use. Itemize steps to build your new design and your selected product confidence test>

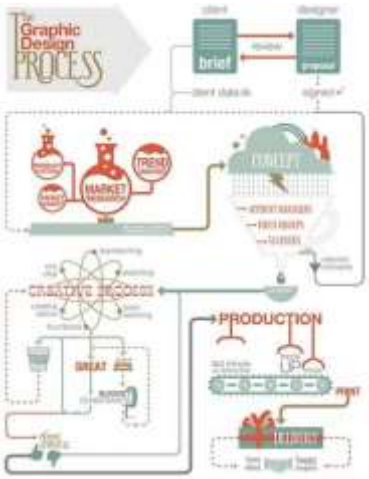


Figure 4. Suggestion of steps that you may want to implement in your product design plan

3.2. Budget

<What is the total budget for the project? Have the funds been already acquired? If not, where is the money coming from? How long will it delay the process? Will it impact the thesis work and/or are there other remedies to the problem? Present in a table is recommended>

3.3. Product Design Reflection, Benefit, Advantages and Limitation

4. Conclusion

<To conclude your work to solve the problem as in the problem statement and achieve the goals stated in the objective. For example: "In this project, techniques for designing an interactive module that used as an educational teaching aid to strengthening learning through play in early childhood education was developed….">

4.1. Product summary

4.2. Product contribution

4.3. Suggestion for future work

References *<Use APA format>*

Report Marking Rubrics

Presentation:

1. The presentation required all team members' participation. All team members will be involved in broadcasting the group's prototype during the presentation session.

2. Prototype examples

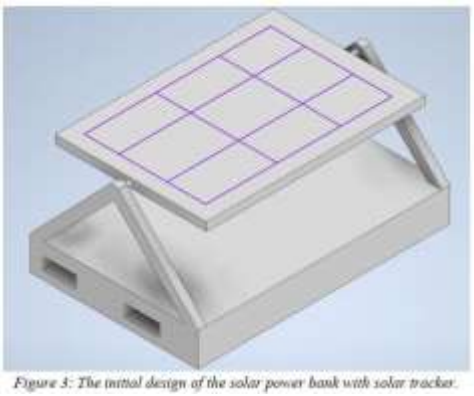


Figure 3: The initial design of the solar power bank with solar tracker.

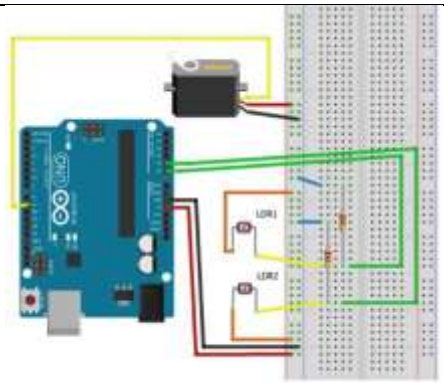
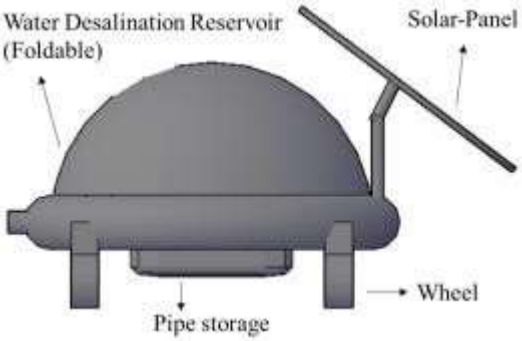


Figure 11: An example circuit connection for a simple solar tracker.







Revolution of earth around the sun



REPORT MARKING RUBRICS

Component Title	EEE316 Engineering Management Project Ratings and Mapping				Percentage (%)	
Criteria	**CLO1-PLO10 (Communication)** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions				**Weight (%)**	**Marks**
	1	**2**	**3**	**4**		
Report Organization	Lack of organization and weak content link. Difficult to understand the main idea.	Content between subtopics is linked in most areas. Able to understand the main idea	Content between subtopics is linked in most areas. Easy to understand the main idea.	The content strong link between subtopics. The content is presented clearly and easy to follow.	2	
Introduction	Weak statement of the main idea, product background and problems. Weak justification of the concept with unclear objectives.	Fairly address the statement of the main idea, product background and problems. The justification of the idea can be understood with unclear objectives.	Address the statement of the main idea, product background and problems sufficiently. The justification of the idea can be understood with objectives.	Clearly highlighted the statement of the main idea, product background and problems. The justification of the idea is presented. The objectives are linked to the problem statement, as well as to the final product.	3	
Formatting and Language	Obvious and significant formatting mistakes. Poor language, bad grammar and vocabulary mistakes	Minor formatting issues. Minor language error, little grammar and sentence construction mistake	Good formatting issues. Language, grammar and sentence constructions of the report are usually correct.	Excellent formatting. The language, grammar and sentence constructions of the report are perfectly written.	3	
Reference and Citation	References are not sufficient with the wrong format. Citations are not organized and cited non-relevant sources to the project.	References are not sufficient with the inconsistent format. Citations are not organized but cited relevant sources to the project.	References are sufficient with old and recent sources. The overall format is consistent. Citations well organized and cited relevant sources to the project.	Sufficient references list of recent years with no formatting errors. Citations are well organized in the texts, and sources are relevant.	2	
TOTAL (PLO10: Communication)					**10**	

Criteria	**CLO3-PLO12 (Life-long Learning)** Recognise the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.				**Weight (%)**	**Marks**
Creative, Innovative and Inventive Ideas	Poorly present innovation and/or invention of the product. Unable to present ideas or specific understanding of the contexts in which creativity applied.	Fairly present innovation and/or invention of the product. Present ideas or specific understanding of the contexts in which creativity is applied sufficiently.	Present the innovation and/or invention of the product clearly. Present ideas or specific understanding of the contexts in which creativity is applied sufficiently.	Excellently present the innovation and/or invention of the product. Present the generation of the ideas or specific understanding of the contexts in which creativity is applied clearly.	3	
Planning and Task Performed	Project plan and process flow are difficult to understand and weakly linked to the main idea. Missing or unclear schedule, poor planning and task delegation.	The project plan and process flow are understandable but weak linked to the main idea. Poor schedule with proper planning and task delegation.	Good project plan and process flow linked correctly to the main idea. Good schedule with good planning and task delegation.	Present project planned and process flow clearly, address the methods and stages precisely. Perfectly delegate tasks amongst the team members.	3	
Project Risk Management	No clear description of risk.	Present the description of the risk partially.	Provide a good description of risk.	Provide an excellent description of risk.	3	
Product Design Reflection, Benefit, Advantages and Limitation	Lack of product design reflection, benefit, advantages and limitation.	Provide product design reflection, benefit, advantages and limitation information.	Presented product design reflection, benefit, advantages and limitation information clearly and well discussed.	Present product design reflection, benefit, advantages and limitation information excellently.	3	
Recommendation and Conclusion	Inappropriate recommendations for further improvement. The final work and objectives conclusion is poor, difficult to understand conclusion and missing future work.	Fairly present the recommendations for further improvement. Final works partially met the original objective, conclusion and future work not clearly described.	Presented good recommendations for further improvement. Final findings met the objectives, product accomplishment concluded nicely, future work reported moderately	Excellent recommendations for further improvement. The final product exceeded expectations, highlighted the product contribution or novelty, and presented the future work.	3	
TOTAL (PLO12: Life-long Learning)					**15**	
TOTAL					**25**	

PRESENTATION MARKING RUBRICS

Component Title	EEE316 Engineering Management Project Ratings and Mapping				Percentage (%)	
Criteria	**CLO1-PLO10 (Communication)** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions				**Weight (%)**	**Marks**
	1	**2**	**3**	**4**		
Content and presentation	Video contents and length are inappropriate, lack of information, fail to include or poor introduction components, product development, product detail design and conclusion properly. Lack of innovation and novelty of the proposed prototype.	Video contents and length are appropriate, reasonably provide enough information, acceptable introduction components, product development, product detail design and conclusion. Partially show advancement from the existing product.	Video contents and length are appropriate. Information is adequate. Presented the introduction components, product development, product detail design and conclusion clearly. Partially show advancement from the existing product.	Video contents and length are interactive and creative, clearly present all the product information, and effectively present all the introduction components, product development, product detail design and conclusion. Show the novelty of the proposed prototype.	4	
Delivery of presentation	Video presentation not delivered comprehensively, the video not uploaded into selected social media, lack of creativity and organization, poor language and video flow. Difficult to understand the proposed teaching aid product presented in the video.	The reasonable video presentation and delivered using selected social media with minimum organization or flow issues. Demonstrate minimum creativity in the video presentation. Acceptable language with unclear pronunciations and easy-to-understand the proposed product's gist.	Good video presentation was delivered using selected social media with minimal organization and flow issues. The video somewhat shows some creativity. Use good language with clear pronunciations, nice video flow and easy to understand the gist of proposed teaching aid product.	Video presentation delivered excellently and highly creatively using selected social media without organization or flow issues. Presentation conducted effectively, excellent pronunciations, nice video flow and easy to understand the gist of the proposed teaching aid product.	8	
Prototype, time management and Q & A	Prototype is not presented during the presentation. Unable to answer the questions and relate with the design concepts. The presentations take > 10 minutes	Only present the developed product's drawing figure or mock prototype during the presentation. Reasonably able to answer the questions and relate with the design concepts. The presentations take > 10 minutes.	Successfully presented the working prototype during the presentation. It is tangible but not really well presented. Able to answer the questions and relate with the design concepts clearly. The presentation session was completed within 10 minutes.	Excellently presented the working prototype during the presentation. It is tangible and well explained. Able to answer the questions clearly, and all team members participated. The presentation session was completed within 10 minutes.	3	
TOTAL					**15**	